SOC Home Lab — Wazuh SIEM Project

Portfolio Summary | RDP Brute-Force Detection & Custom Correlation Engineering

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Resume Bullet (ready to paste)

Built a SOC home lab using Wazuh SIEM across a 3-VM environment (Ubuntu manager, Windows 11 agent, Kali attacker); authored a custom frequency-based correlation rule (5+ failed RDP logons within 120s) mapped to MITRE ATT&CK T1110, validated against a live Hydra brute-force attack; diagnosed and documented a Wazuh Active Response platform limitation related to Windows EventChannel log collection.

What I Built

- Deployed a full open-source SIEM stack (Wazuh Indexer, Manager, Dashboard) on Ubuntu Server from scratch.
- Onboarded a Windows 11 endpoint as a monitored agent and verified live log ingestion end to end.
- Simulated a real attacker technique (MITRE ATT&CK T1110 – Brute Force) using Hydra from a Kali Linux attack box over RDP.
- Confirmed detection via Wazuh's built-in ruleset, then went further by authoring an original correlation rule in custom XML.
- Validated the custom rule live against repeated attack runs, confirming consistent, accurate detection at the correct severity and MITRE tag.
- Configured Active Response for automated IP blocking; when it failed, root-caused the exact platform limitation via log analysis and community research rather than giving up or guessing.
- Documented the entire investigation — architecture, detection logic, troubleshooting, and MITRE mapping — in a structured report suitable for a SOC analyst work sample.

Core Skills Demonstrated

SIEM deployment & administration · Log collection & normalization · Wazuh custom rule authoring (XML, frequency/timeframe correlation) · MITRE ATT&CK mapping · Offensive testing (Hydra, brute-force simulation) · Windows Security auditing & Event Log analysis · Linux/Windows cross-platform troubleshooting · Incident documentation & reporting

Tech Stack

Wazuh 4.14.8 · Ubuntu Server 26.04 LTS · Windows 11 Pro · Kali Linux · Hydra · UTM / QEMU (Apple Silicon virtualization)

Links

GitHub repository: *[add your repo link here]*
Write-up (Medium): *[add your Medium link here]*

This lab was built entirely on isolated, self-owned virtual machines for educational purposes.